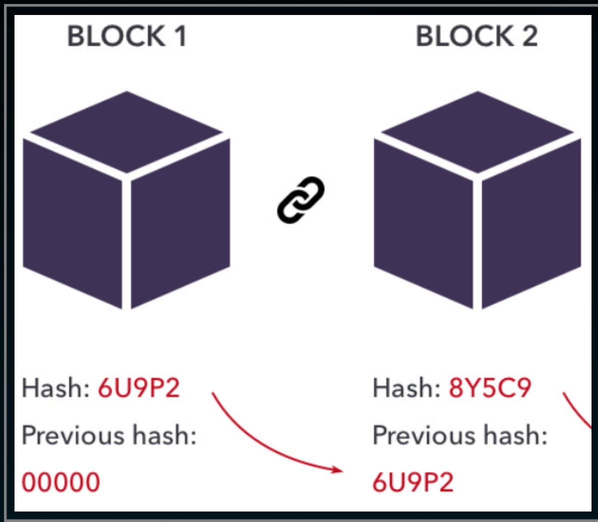
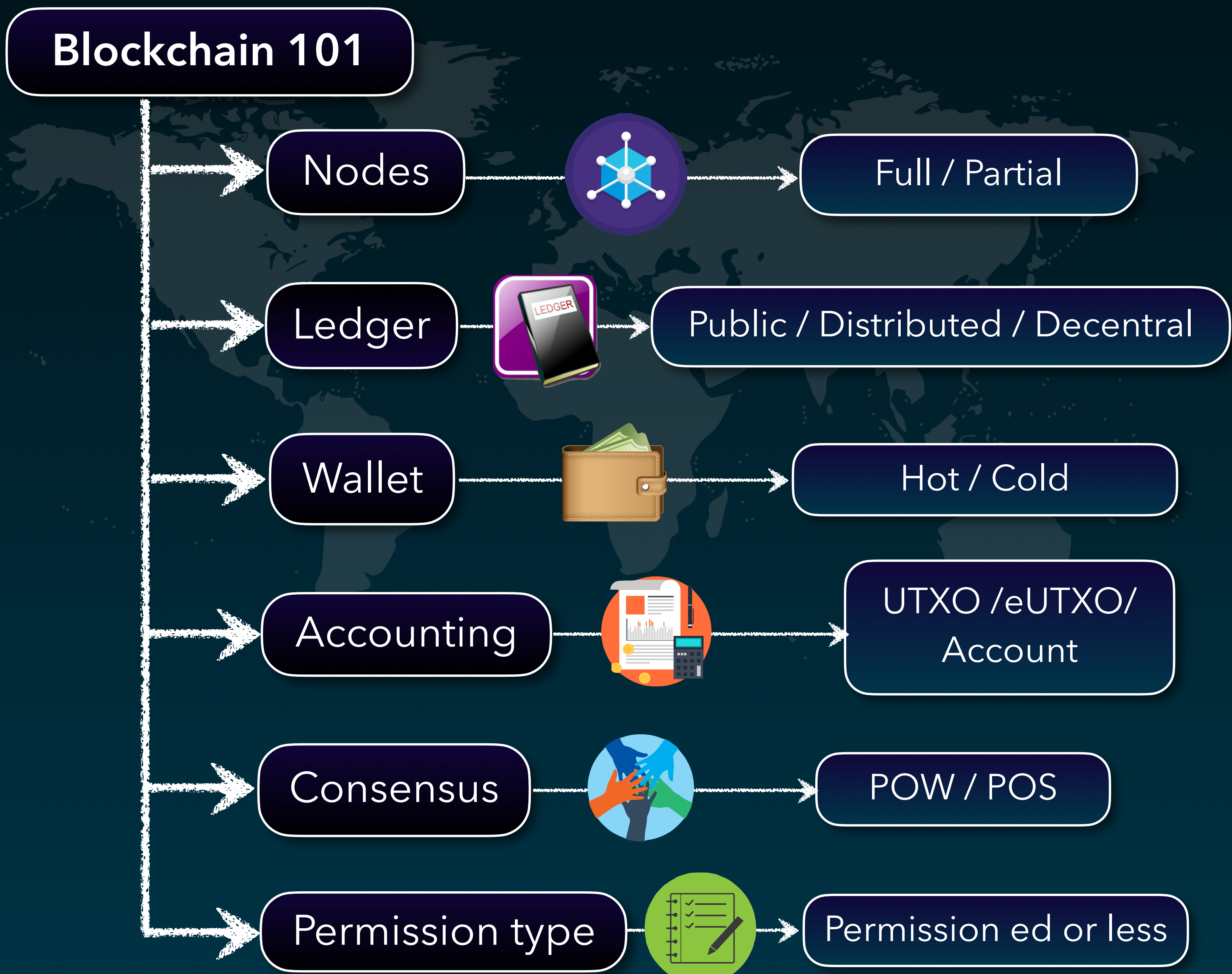
Aarish Bhesania

Blockchain 101

- Blockchain is a ***distributed database*** or ***ledger*** that is shared among the nodes of a computer.
- ***Data*** is ***stored*** in ***Blocks*** each ***block holds hash of previous block*** it is connected in a chain like manner using hash.
- Data in blocks ***cannot*** be ***changed*** only ***new blocks can be added*** if all parties give their ***Consensus*** (Agree).
- Below are the main components of a blockchain.





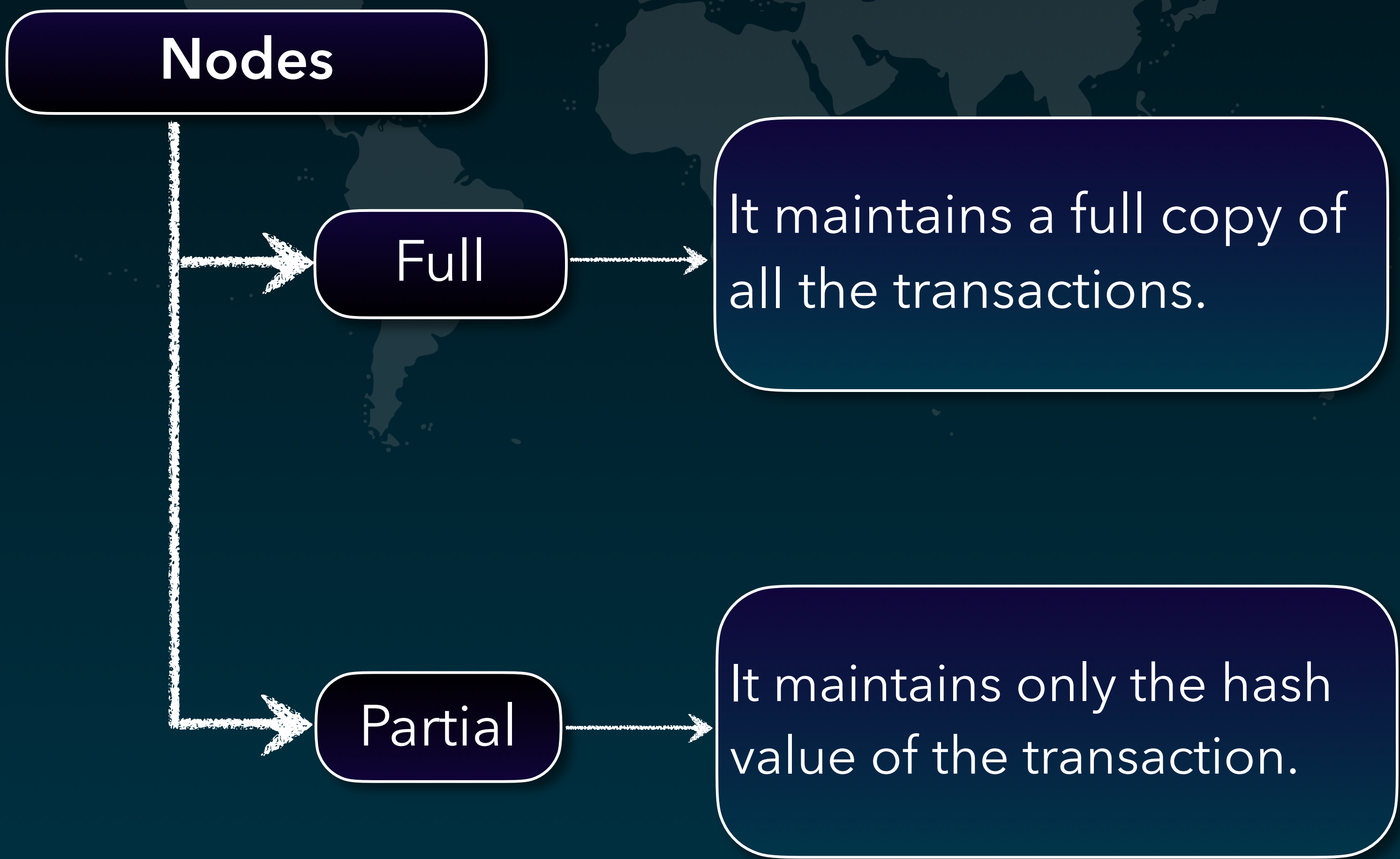
Nodes

✦Full Node

- It maintains a ***full copy of all the transactions.***
- It has the capacity to ***validate, accept and reject*** the transactions.

✦Partial Node

- It is also called a ***Lightweight Node*** because it ***doesn't maintain the whole copy*** of the blockchain ledger.
- It ***maintains*** only the ***hash value*** of the transaction, can be a ***validation Node.***
- The whole transaction is accessed using this hash value only.
- These nodes have ***low storage*** and ***low computational*** power.



Aarish Bhesania

29 OCT 2022

Ledger

✦It is a ***digital database*** of information, their are 3 types of ledgers

✦Public Ledger

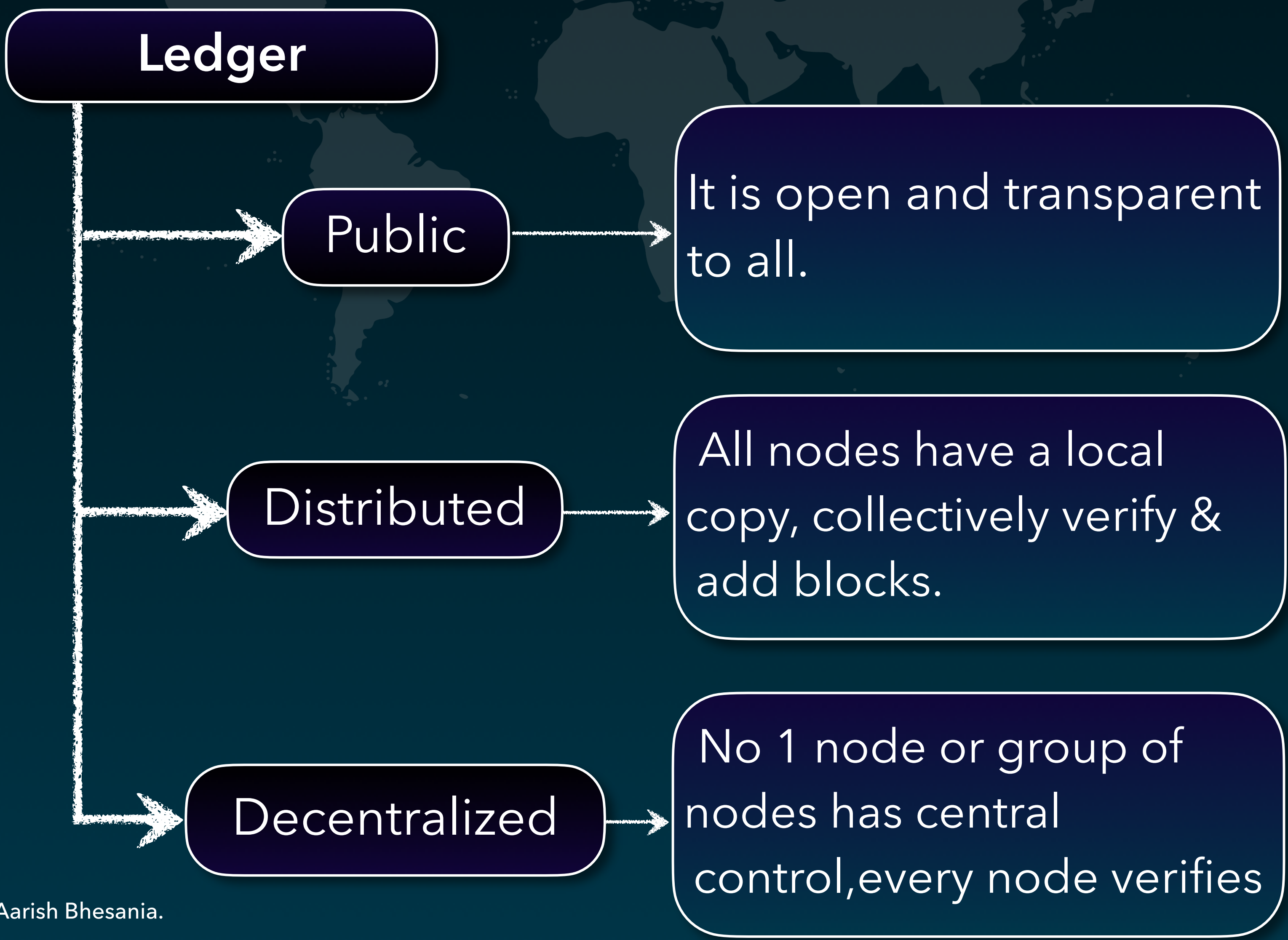
- It is ***open*** and ***transparent*** to all. Anyone in the network can read or write .

✦Distributed Ledger

- In this ledger, all ***nodes have a local copy of the database*** , group of nodes collectively execute the job i.e verify transactions, add blocks in the blockchain.

✦Decentralized Ledger

- In this ledger, ***no one node or group of nodes has a central control***. Every node participates in the execution of the job.



Aarish Bhesania.

Aarish Bhesania

29 OCT 2022

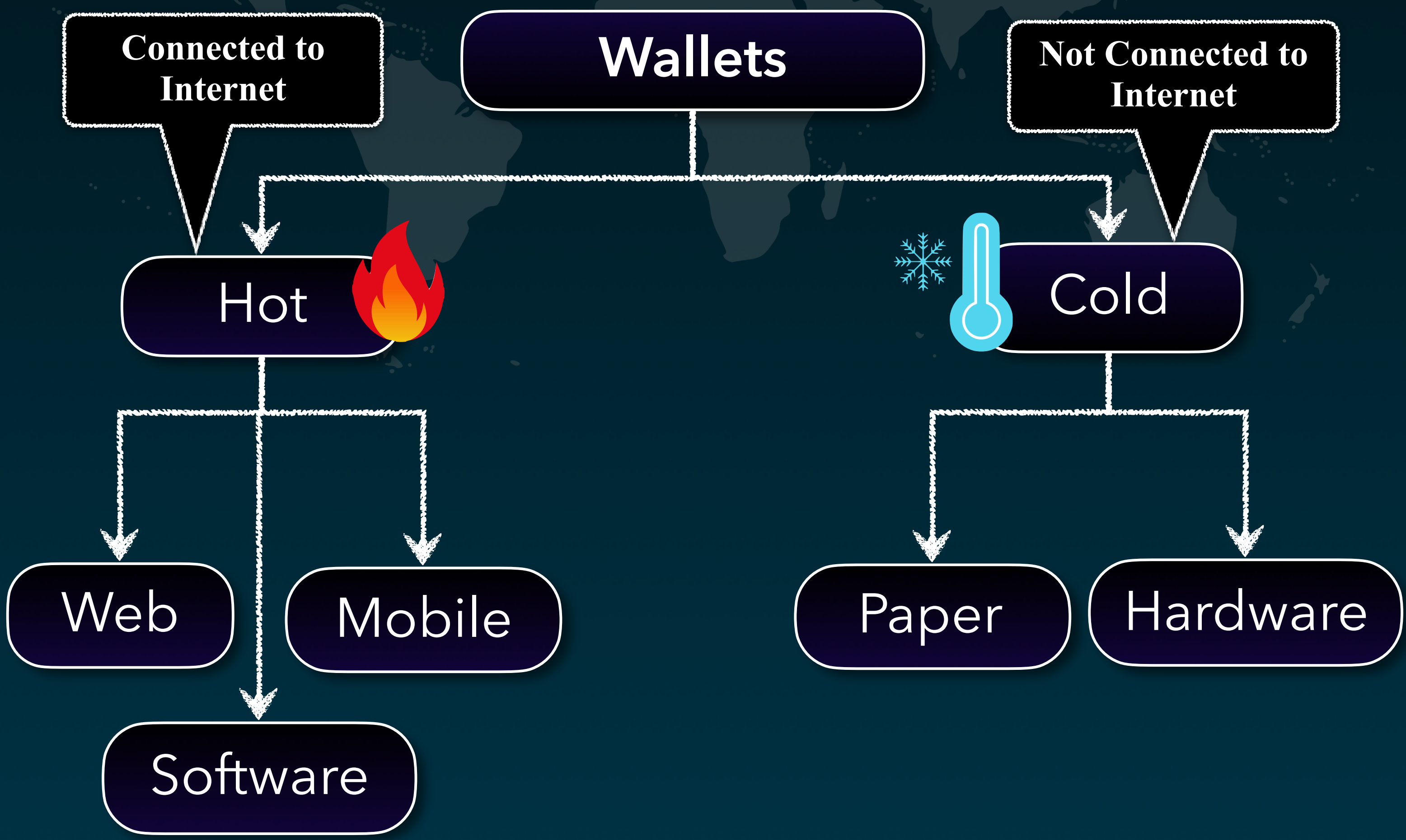
Wallets

✦Wallet allows user to ***store their cryptocurrency***, Every node in the network has a Wallet.

✦ Privacy of a wallet in a blockchain network is maintained using public and private key pairs.

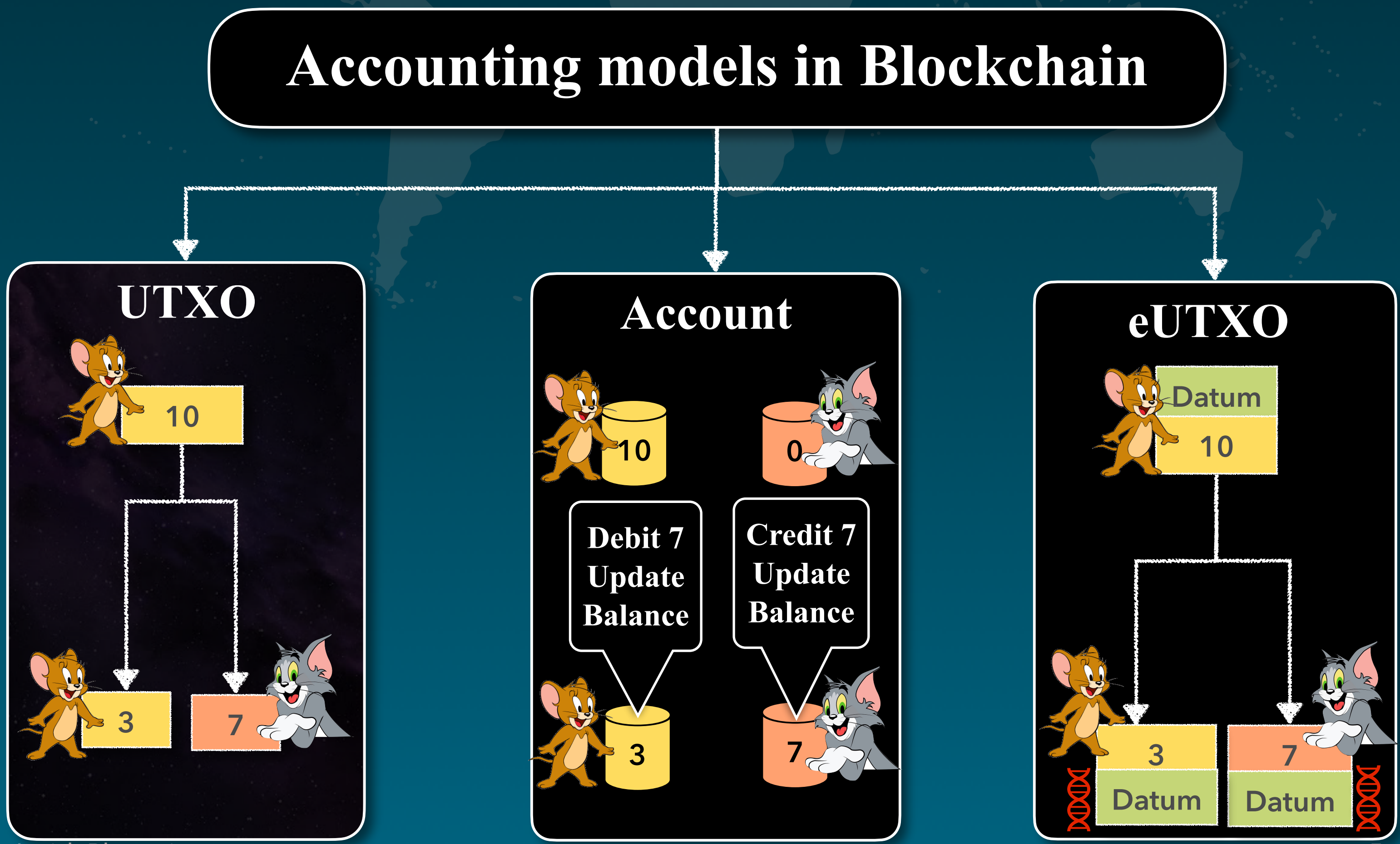
✦Types of wallets

- **Hot Wallet** - are used for online day-to-day transactions and are ***connected to the internet***
  - Types of Hot wallet (***Web , Software , Mobile based***)
- **Cold Wallet** - are ***not connected to the internet***
  - ***Paper based*** -Piece of paper contains the address. The private key is in QR code format.
  - ***Hardware based*** -Physical electronic device that uses a random number generator that is associated with the wallet.



Accounting

- For digital money to be useful, it ***needs to be transferable***.
- The transfer of money on a blockchain is initiated by the owner, creating a transaction. This transaction informs the network about ***how much money is changing hands and who the new owner is.***
- This transfer mechanisms is done by adopting any of the below 3 accounting models.
  - ***UTXO*** - Unspent Transaction Output, the ***change which is left after execution*** of a crypto transaction, is stored in a separate new token.
  - ***Account/Balance based*** - ***Stores assets within accounts*** , these account balances are ***debited , credited as per activity.***
  - ***eUTXO*** **-** Extended Unspent Transaction Output, same as UTXO model additionally it allows addition of ***user data in JSON like format this data is know as Datum***.



Aarish Bhesania

29 OCT 2022

Consensus

✦ A consensus mechanism refers to methodologies ***used to achieve agreement***, trust, and security across a decentralized computer network.

✦ Publicly shared ledgers need a ***mechanism to ensure that all the transactions occurring on the network are genuine and all participants agree*** on a consensus on the status of the ledger, this is know as consensus

✦ It is a ***set of rules that decides on the legitimacy of changes*** made by the various participants (i.e., nodes or transactors) of the blockchain.

✦ Their are many Consensus mechanism but POW & POS are the most common ones.

◉ **POW Proof of Work** - Miners ***compete to complete transactions*** become (validators) on the network in exchange for a reward for their speed and accuracy. ***Used in BITCOIN***.

◉ **POS Proof of Stake** - Miners ***stake their coins on the blockchain*** , in exchange it allows them to ***become validators for validating new blocks*** , used in ***Ethereum*** , ***Cardano*** etc

Consensus

POW → one party (the *prover*) proves to others (the *verifiers*) that a certain amount of a specific computational effort has been expended & new block can be added

POS → Validator is chosen randomly, based on how many coins they staked. The coins act as collateral and after validating a transaction, they receive a reward.

Aarish Bhesania.

Permission Type

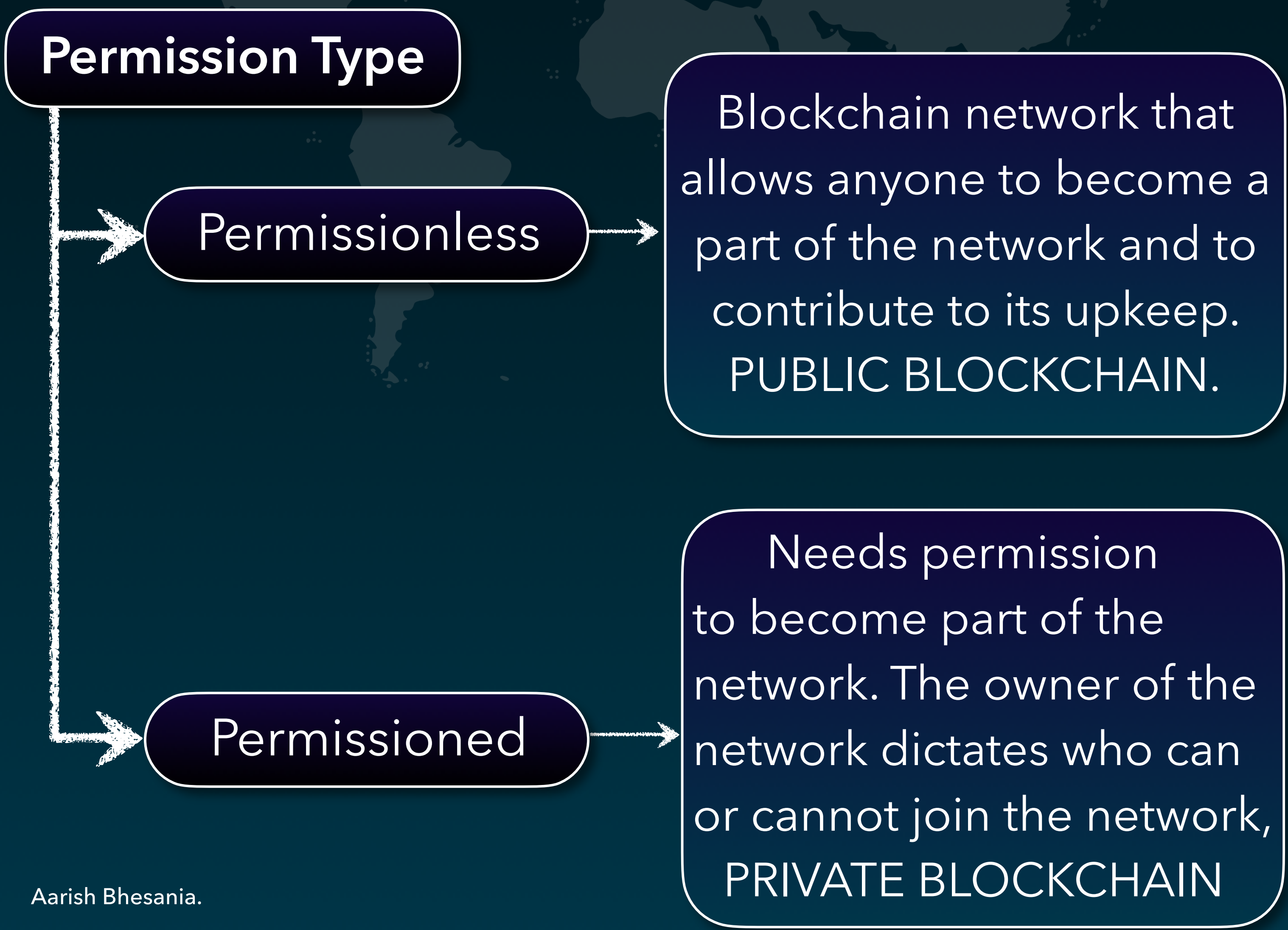
✦ Two fundamentally different blockchain models have emerged: permissioned and permissionless blockchains.

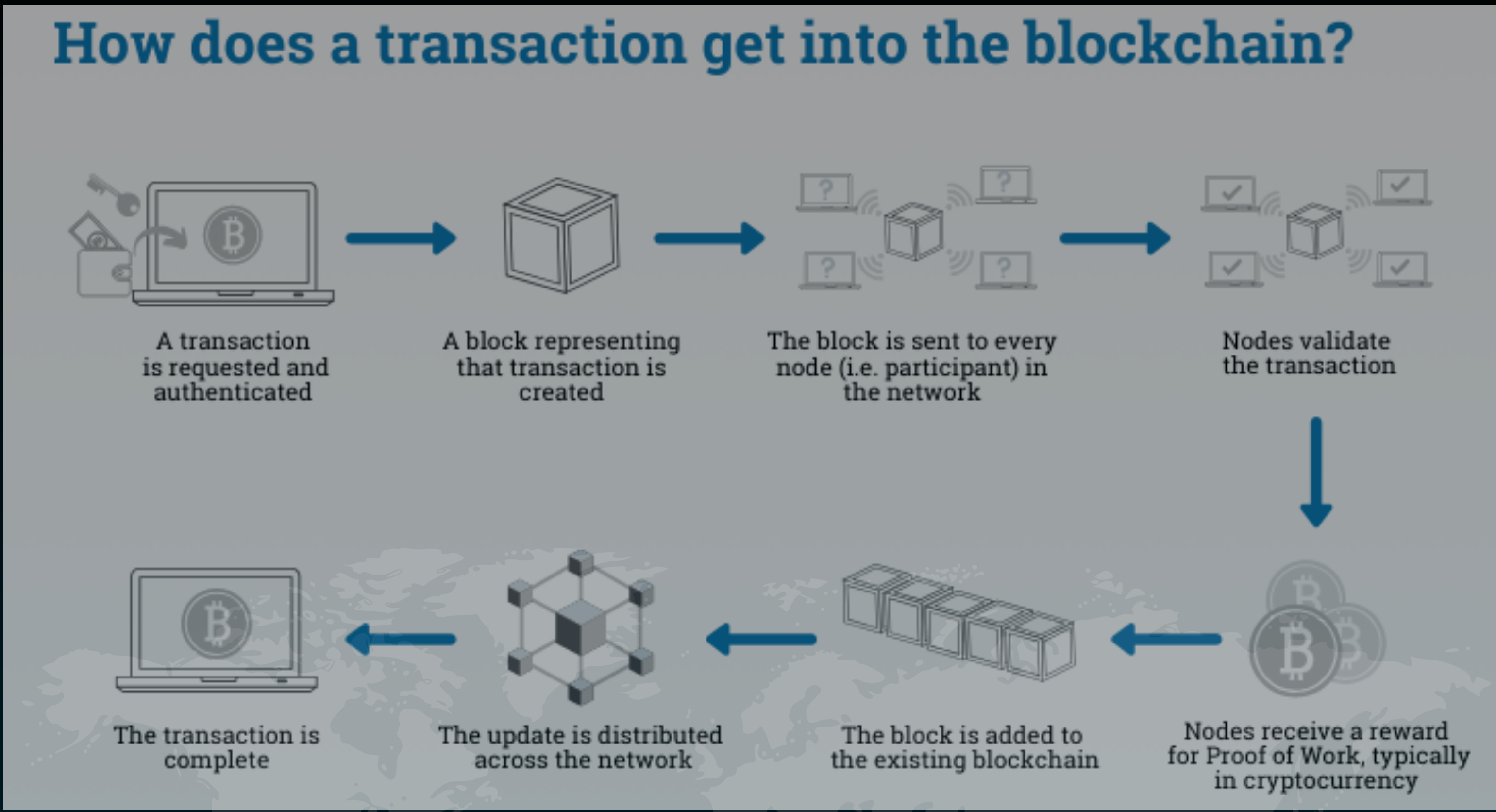
✦ **Permissionless** - ***Public Blockchain***

- Permissionless blockchain is a type of blockchain network that ***allows anyone to become a part of the network and to contribute to its upkeep.***
- Permissionless blockchain is a decentralized ***ledger that is open to the public.***

✦ **Permissioned Blockchains** - ***Private Blockchain***

- In a permissioned blockchain network, ***one needs permission to become part of the network. The owner of the network dictates who can or cannot join the network.***





The End.

Please fell free to suggest & add more on it,
For more articles about payments please follow,

https://www.linkedin.com/in/aarishbhesania

*Special thanks to **Ashutosh Dubey** for Valuable insights & review.*